Position predictions and their limits

LEARNING TARGET · R3 I can predict a position and state its uncertainty.

SECTION A KEY VOCABULARY

Start x_0	Position when the chosen interval begins.
Bound	An allowed range, not a standard deviation.
Half-width	The ± value; full width is twice as large.
Assumption	A condition treated as true, such as steady v.

SECTION B KEY EQUATIONS + UNITS

$$x_f = x_0 + vt$$

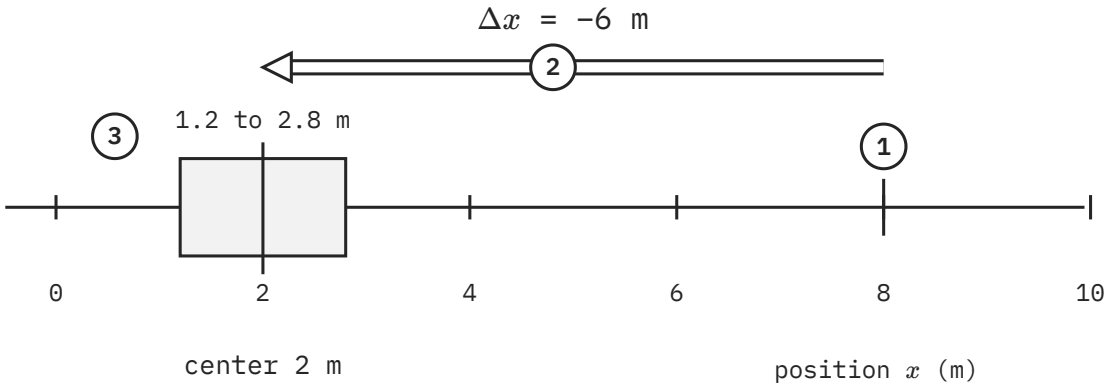
$$\delta x_f = \delta x_0 + t\delta v$$

t exact time, s
δx bound, m

SECTION C COLOR IT · A BOUNDED FORECAST

Center plus range.

$x_0 = 8 \pm 0.5$ m $\qquad$ $v = -2.0 \pm 0.1$ m/s; $t = 3$ s



☐ **1 · BLUE** Start and end positions

☐ **2 · BLUE** Displacement

☐ **3 · BLUE** Prediction interval

SECTION D WORKED EXAMPLE

A cart starts at **8 ± 0.5 m** and moves at **−2.0 ± 0.1 m/s** for an exact **3 s**. **Find its predicted position interval.**

GIVEN	EQUATION	WORK	ANSWER
x_0 = **8 ± 0.5 m** v = **−2.0 ± 0.1 m/s** t = **3 s**; steady v	Steady v: $x_f = x_0 + vt$. Add input bounds.	$x_f = 8 - 2(3)$ $\delta x = .5 + 3(.1)$ = **0.8 m**	**1.2 to 2.8 m** **2.0 ± 0.8 m**; full width **1.6 m**.

SECTION E THE 2026-XJ THREAD

DAY 5 OF 22 · NEW TOOL: BOUNDED FORECAST

XJ's position and velocity readings have limits. A forecast should show both the central result and the interval allowed by those limits; the interval does not give an impact probability.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else